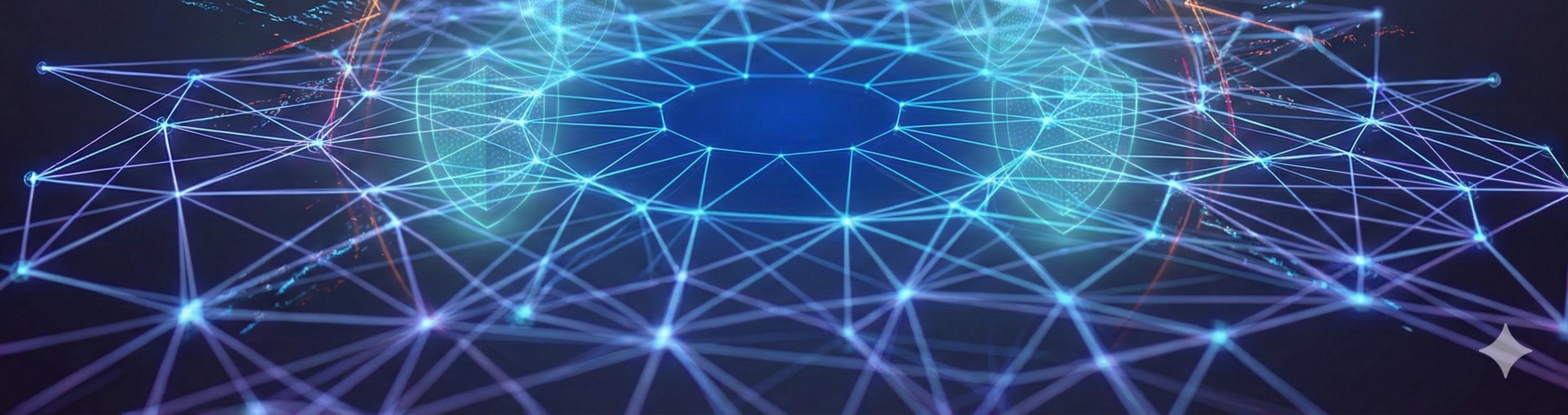
MEC Defense: Securing Smart Stadium Critical Infrastructure

Trabalho realizado por:
José Fernandes, nº114472
Diogo Ferreira, nº114002
Rúben Franco, nº114827
Diogo Carvalho, nº113221
Gonçalo Martins, nº112678
Tiago Rocha, nº120515

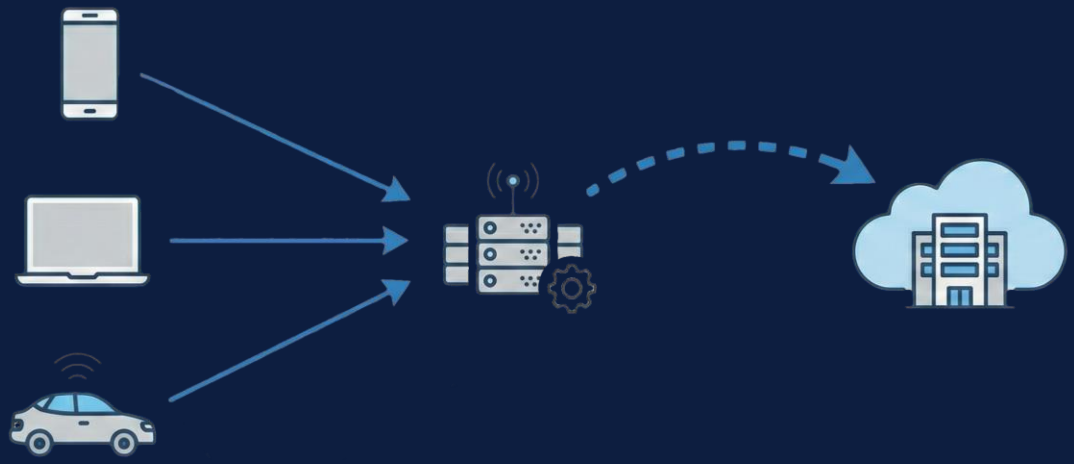


Context

- Modern stadiums evolve into massive "Smart Environments"
- Deployed mobile networks provide internet to fans, but also critical infrastructure
- Data is processed locally on a MEC platform to reduce latency
- The system should be able to take real-time decisions based on multiple factors
- Public and external traffic should have little to no effect on the sensitive AI processing

Problems

- Shared infrastructure for different workloads
- Security blind spot, requiring careful traffic segregation
- External threats may also pose a risk to the internal network



Goals and Expected Results

- Implementation of a Security Operations Center (SOC)
- Recognize and log atacks (e.g. DDoS attacks)
- Automate defensive mechanisms
- Traffic segmentation and isolation
- Take real-time decisions based on multiple factors

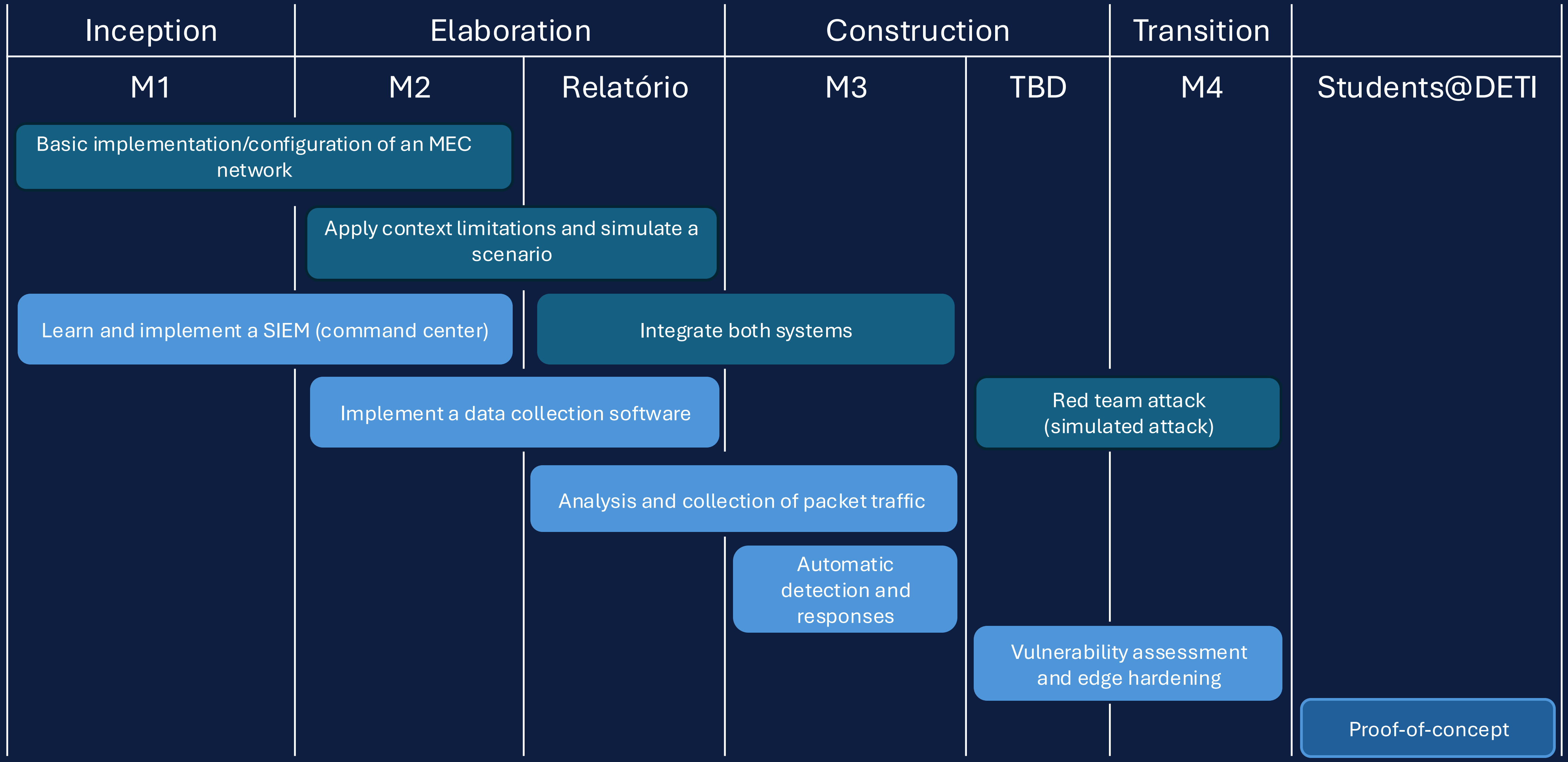
Technologies

- Docker/Kubernetes
- OSMANO
- Kafka
- SIEM/Threat Intelligence: Wazuh,

ELK Stack (Elastic Security), or Splunk

- Telemetry and Observability: Prometheus
- Network Traffic Analysis: Suricata, Zeek
- Packet Monitoring and Filtering: Cilium
- Protocols: IPsec, WireGuard, TLS 1.3,

ModSecurity (WAF).


Inception
Elaboration
Construction
Transition
M1
M2
Relatório
M3
TBD
M4
Students@DETI
Basic implementation/configuration of an MEC network
Apply context limitations and simulate a scenario
Learn and implement a SIEM (command center)
Integrate both systems
Implement a data collection software
Red team attack (simulated attack)
Analysis and collection of packet traffic
Automatic detection and responses
Vulnerability assessment and edge hardening
Proof-of-concept

Communication Plan

- Initial communication between group member done in Discord
- Communication with supervisors through Slack
- Software components managed through GitHub